WORKED SOLUTION

STEP III 2001, Question 3

Pure • Roots of Quadratics

STEP III 2001, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s3-q3

The key idea

Translate statements about the roots into statements about the coefficients. With roots α and β we have $\alpha + \beta = b$ and $\alpha\beta = c$, and the discriminant tells us when the roots are real. For part (ii) it's easier to think about the graph of $f(x) = x^2 - bx + c$: both roots lie strictly between -1 and 1 exactly when the parabola is positive at ± 1 and its vertex lies between them.

Solution

Throughout, write $f(x) = x^2 - bx + c$, and when the roots are real call them $\alpha \leq \beta$, so that

$$\alpha + \beta = b, \qquad \alpha\beta = c, \qquad \alpha, \beta = \tfrac{1}{2}(b \mp \sqrt{b^2 - 4c}).$$

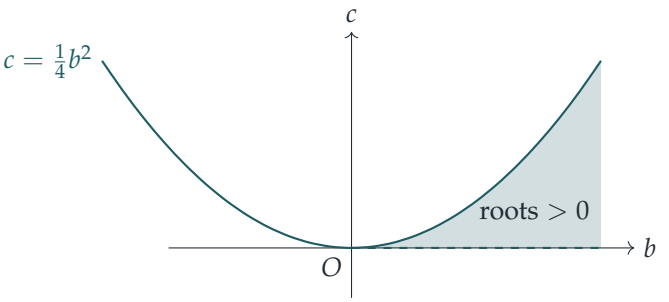
A repeated root counts as two equal real roots.

Part (i)

($\Rightarrow$) Suppose the roots are real and positive. Real roots need $b^2 - 4c \geq 0$. Then $b = \alpha + \beta > 0$ and $c = \alpha\beta > 0$, since both roots are positive. Together these give $b > 0$ and $b^2 \geq 4c > 0$.

($\Leftarrow$) Suppose $b > 0$ and $b^2 \geq 4c > 0$. Since $b^2 - 4c \geq 0$, the roots are real. Since $\alpha\beta = c > 0$, they are non-zero and have the same sign. Since $\alpha + \beta = b > 0$, that common sign can't be negative. So both roots are positive.

This proves the equivalence. In the b-c plane the conditions say $b > 0$ and $0 < c \leq \frac{1}{4}b^2$: the region in the first quadrant between the positive b-axis and the parabola $c = \frac{1}{4}b^2$. The parabola is included (repeated root $\frac{1}{2}b > 0$), and the b-axis is excluded (there $c = 0$ and one root is 0).



The solid curve (for $b > 0$) belongs to the region; the dashed segment of the b-axis and the origin do not.

Part (ii)

We need both roots real with $-1 < \alpha \leq \beta < 1$. We claim that, given real roots, this happens if and only if

$$f(1) = 1 - b + c > 0, \qquad f(-1) = 1 + b + c > 0, \qquad -1 < \tfrac{1}{2}b < 1.$$

($\Rightarrow$) If $-1 < \alpha \leq \beta < 1$ then $f(1) = (1-\alpha)(1-\beta) > 0$ and $f(-1) = (1+\alpha)(1+\beta) > 0$, as each bracket is positive. Also $\frac{1}{2}b = \frac{1}{2}(\alpha+\beta)$ is the average of two numbers in $(-1,1)$, so it lies in $(-1,1)$.

($\Leftarrow$) Because $f(x) = (x-\alpha)(x-\beta)$ is ≤ 0 exactly on $[\alpha,\beta]$, the condition $f(1) > 0$ says that 1 is not in $[\alpha,\beta]$. So either $1 < \alpha$ or $1 > \beta$. The vertex $\frac{1}{2}b$ lies in $[\alpha,\beta]$, and $\frac{1}{2}b < 1$, so $\alpha \leq \frac{1}{2}b < 1$. That rules out $1 < \alpha$, so $\beta < 1$. In the same way, $f(-1) > 0$ and $\frac{1}{2}b > -1$ give $\alpha > -1$.

? **Ask yourself** Why can't we just use $f(1) > 0$ and $f(-1) > 0$ on their own?

Both roots could lie on the same side of the interval. For example, $x^2 - 5x + 6 = 0$ has roots 2 and 3, yet $f(\pm 1) > 0$. The vertex condition $|b| < 2$ is what puts the roots between -1 and 1 rather than both beyond one end.

Combining these with the reality condition $b^2 \geq 4c$, the region is

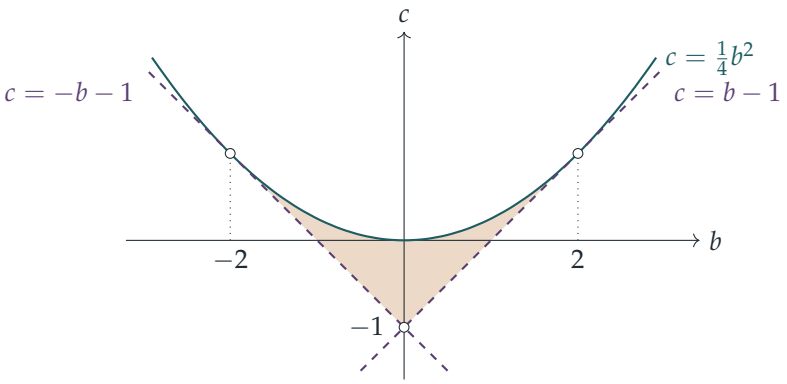
$$|b| < 2, \qquad c > b - 1, \qquad c > -b - 1, \qquad c \leq \tfrac{1}{4}b^2,$$

that is, $|b| < 2$ and $|b| - 1 < c \leq \frac{1}{4}b^2$.

To sketch it, note how the lines meet the parabola. Since

$$\tfrac{1}{4}b^2 - (b-1) = \tfrac{1}{4}(b-2)^2 \geq 0,$$

the line $c = b - 1$ lies below the parabola and touches it only at $(2,1)$. So it is the tangent there. By symmetry ($b \mapsto -b$), $c = -b - 1$ is the tangent at $(-2,1)$. The two lines cross at $(0,-1)$. The region is the curvilinear triangle bounded by these two tangents and the arc of the parabola between the points of contact.



The arc of the parabola with $|b| < 2$ is included (a repeated root $\frac{1}{2}b$ with $|\frac{1}{2}b| < 1$). The two dashed tangent segments are excluded, because on them one root is ± 1. The three corners $(\pm 2, 1)$ and $(0, -1)$ are excluded too.

Checks

- $(b, c) = (0, -\frac{1}{4})$ is inside the part (ii) region: the roots are $\pm\frac{1}{2}$. $(b, c) = (0, -1)$ is the excluded corner: the roots are exactly ± 1.
- $(b, c) = (2.7, 1.8)$ satisfies $c > |b| - 1$ and $c \leq \frac{1}{4}b^2$ but has $|b| > 2$. Its roots are 1.2 and 1.5, so it is correctly left out by the vertex condition.
- We tested both regions numerically, comparing the exact roots with the stated conditions at over 115 000 rational points (b, c), including points on every boundary curve. There were no disagreements.

Common slips

- Forgetting the vertex condition $|b| < 2$ in part (ii), which wrongly adds the thin slivers between each tangent line and the parabola beyond $|b| = 2$.
- Getting the boundaries wrong: the parabola is included in both parts, but the b-axis in (i) and the tangent lines in (ii) are not.

Beyond STEP

The conditions on $f(\pm 1)$ and the vertex are a small case of a general question: when do all roots of a polynomial lie in a given region? For the unit disc this is answered by the Schur–Cohn test. For the left half-plane it is the Routh–Hurwitz criterion, which control engineers use to decide whether a system is stable. Part (ii) is exactly the stability region for the recurrence $u_{n+2} = bu_{n+1} - cu_n$ with real characteristic roots. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

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